

DOES STORED-TOKEN UNCERTAINTY IMPROVE TRAJECTORY REPAIR? A CONTROLLED HOTPOTQA STUDY

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ABSTRACT

When a language model agent fails, recovery can preserve a trajectory prefix and regenerate its suffix, or restart from the original task. We test whether stored-token uncertainty helps select a recovery origin beyond simple positional choices in a ReAct-style agent. A controlled experiment with Qwen2.5-32B-Instruct-AWQ freezes 250 HotpotQA questions before inference. Its 113 initial failures receive six repair conditions and three seeds under matched retry hints, generated-token caps and new-step allowances. The prespecified perplexity/argmax policy with two-step backtracking achieves 9.73% repair success, versus 11.21% for full restart: a difference of -1.47 percentage points, with a question-level 95% bootstrap interval from -4.13 to +0.88. Its success rate equals the position-matched random control; none of the four prespecified comparisons shows a clear advantage after Holm correction. Exploratory trace diagnostics show that the treatment differs from restart on only 43 failed questions and uses 29.5% more prompt tokens, with similar generated-token use. All 1,058 unique repair executions pass independent replay, scoring and allowance audits. Historical exclusion IDs are reconstructed, with the original full pool comparison unavailable. A separate 250-question cohort compares restart, uncertainty, and same-model diagnosis/replay at three allowances, with 2,079 additional unique repairs. Uncertainty again shows no clear advantage in six Holm-adjusted comparisons. These results do not establish equivalence or an advantage under equal total compute.

1 INTRODUCTION

ReAct agents interleave reasoning, tool calls, and observations (Yao et al., 2023); a failed answer can reflect mistakes at several points. Recovery must decide both what to change and how much context to retain. Restart discards potentially useful work. Suffix regeneration preserves that work but constrains recovery to the retained reasoning and observations.

Error attribution and recovery need not have the same objective. Attribution asks which step is erroneous, whereas recovery asks where a new rollout is most likely to succeed at an acceptable cost. A judge may identify an incorrect conclusion that is downstream of incomplete evidence gathering. Conversely, a locally uncertain action can be useful exploration. Neither uncertainty nor agreement with an error annotation directly establishes the value of intervening at that step.

We examine this distinction using a common execution loop for targeted regeneration and restart. The historical framework crosses five signals with six origin-selection rules and uses a larger model for privileged reference diagnostics. The completed six-condition study freezes one stored-token policy and evaluates six conditions under matched recovery allowances on HotpotQA. It uses no judge or outcome-based policy search. Reference annotations are not treated as infallible causal labels. Prior work already studies intervention-based debugging, localized QA repair with prefix reuse, and repair versus resampling (Ma et al., 2026; Jiao et al., 2026; Luan et al., 2026). The unresolved question here is whether stored-token uncertainty adds origin-selection value beyond position alone, without a separate diagnosis model, in a fixed tool environment. Neither prefix reuse nor the distinction between attribution and recovery is claimed as new.

The evidence does not support a universal short-trajectory repair versus long-trajectory restart rule: evidence quality, error types, dataset, and budget can influence the ordering. We focus on three contributions:

1. A prespecified six-condition HotpotQA comparison with question-level paired inference, including a development-fitted position control and a no-backtracking ablation; no clear uncertainty advantage is observed. A separate 250-question follow-up adds a same-model diagnosis/replay adaptation at three recovery-token allowances, also without a clear uncertainty advantage.
2. An exploratory decomposition of shared executions and observed resource use, which exposes the limited number of distinct interventions and separates output-token allowances from repeated prompt processing.
3. Audited replay, reference scoring, complete trial coverage and frozen inputs supporting reproduction of the controlled comparison. Older three-dataset summaries are retained as separate historical context.

2 RELATED WORK

Agent recovery and self-correction. Reflexion uses verbal feedback from previous episodes to inform subsequent attempts (Shinn et al., 2023), while Self-Refine iteratively improves outputs using generated feedback (Madaan et al., 2023). Huang et al. (2024) document limitations of intrinsic reasoning self-correction without external feedback. Our benchmark provides an external correctness gate that selects failed attempts. Conditional on that gate, we study suffix origins, not online failure detection.

Localized repair and replay. ReAgent uses local and global rollback in multi-agent QA (Xinjie et al., 2025). Doctor-RAG diagnoses failure type and location, then reuses prefixes and evidence for local repair on HotpotQA, MuSiQue, and 2WikiMultiHopQA (Jiao et al., 2026). This is direct task-level overlap, although its trained diagnoser, retrieval setting, and repair operators differ from our stored-token selector. DoVer evaluates intervention-driven debugging through task recovery (Ma et al., 2026). SymTrace reconstructs recorded prefixes to examine repair versus resampling (Luan et al., 2026); AgentRewind coordinates context and workspace rollback with memory of prior attempts (Zhuang et al., 2026). Our offline tool replay is narrower than general environment rollback.

Step evaluation and failure attribution. Process supervision evaluates intermediate reasoning (Lightman et al., 2024), and Who&When studies multi-agent failure locations (Zhang et al., 2025). AgentRx uses executable constraints and violation logs to support diagnosis (Barke et al., 2026). REFLECT, Causal Agent Replay, and CausalFlow study interventions for attribution or repair (Lin et al., 2026; Shah, 2026; Bonagiri et al., 2026). AgenticRAG-FP injects faults and re-executes downstream reasoning to test attribution against known interventions (Pothuru, 2026). Our naturally failed traces lack such intervention-grounded causal labels.

Uncertainty signals. Self-consistency aggregates sampled reasoning outputs (Wang et al., 2023), and verbalized confidence can provide useful calibration information (Tian et al., 2023). ERGO already uses entropy changes for conversation resets (Mohammad Khalid et al., 2025). The archived framework includes action-level signals and stored-token statistics; the controlled study tests perplexity alone, without assuming calibration as error probabilities. Appendix H specifies the closest comparisons and information differences. No head-to-head advantage over these methods is established.

3 PROBLEM AND EVALUATION TARGETS

An instance consists of a question x , an offline document collection D_x , and a reference answer y . An agent produces a trajectory $\tau = (s_0, \dots, s_{T-1})$. Each step contains generated reasoning, a tool action, its argument, and the returned observation. The available actions are `search`, `lookup`, and `finish`.

Failure gate. The implemented QA success criterion accepts an answer when its normalized exact match is one or its token F1 is at least 0.5. This threshold is a study-specific recovery criterion, not the official aggregate metric for every benchmark. Only trajectories that fail this gate enter repair. Evaluation therefore estimates recovery conditional on a known benchmark failure; the reference answer is not available to an ordinary deployed failure detector.

Repair origin. For an origin $k \in \{0, \dots, T-1\}$, targeted repair retains $s_0, \dots, s_{k-1}$ and generates a new suffix. Tool calls in the retained prefix are replayed to restore the environment state. Full restart uses $k = 0$. With backtracking offset b , the effective origin is

$$k' = \max(0, k - b). \quad (1)$$

Backtracking can make several nominal strategies share the same origin, and can turn targeted repair into a restart. Consequently, strategy counts are not counts of independent interventions.

Reference annotations. Let k_J denote the earliest erroneous step identified by a language model judge. Agreement with k_J measures reference-label localization. It is distinct from the origin that maximizes expected repair success,

$$k^*(\tau, B) = \arg \max_k \Pr[\text{correct}(\text{repair}(\tau, k, B))]. \quad (2)$$

The latter depends on the agent, retained context, recovery prompt, and budget B , and is not observed merely by asking a judge to label a trace.

Success across seeds. For failed question q , strategy a , and generation seed r , write the binary repair outcome as Y_{qar} . Our primary descriptive statistic is

$$\hat{R}_a = \frac{1}{N} \sum_{q=1}^N \frac{1}{S_q} \sum_{r=1}^{S_q} Y_{qar}. \quad (3)$$

Each question receives equal weight. Thresholding each question’s mean success to obtain a majority-over-seeds outcome measures repeatability, not the success rate of one sampled repair, and must be reported separately.

4 UNCERTAINTY-GUIDED REGENERATION

4.1 PRESPECIFIED STORED-TOKEN POLICY

The controlled study uses step perplexity from stored sampled-token log probabilities. For log probability ℓ_{it} of generated token t in step i , define

$$U_i^{\text{ppl}} = \exp \left(\min \left\{ 20, -\frac{1}{n_i} \sum_{t=1}^{n_i} \ell_{it} \right\} \right). \quad (4)$$

The policy selects the highest finite step score, breaking ties toward the earlier step, then subtracts two trajectory indices and clamps at zero. No additional uncertainty-acquisition model calls are required. Unusable measurements are skipped; an empty usable profile falls back to origin zero. This stored-token measure requires no additional sampled rollouts or diagnosis model. The policy was prespecified before main-study outcomes, rather than chosen as the best-performing development or test variant; this is not an evaluation of the best possible uncertainty estimator.

The historical framework also evaluates entropy, sampled-token probability complement, action disagreement and verbalized uncertainty with several localization rules. Appendix A preserves their definitions. Those historical configurations are not additional conditions in the controlled experiment.

4.2 INTERVENTIONS AND INFORMATION ACCESS

Full restart discards the failed context; targeted policies retain a prefix. All six controlled conditions receive the same generic retry hint. The uncertainty selector can inspect the complete failed trace’s

uncertainty profile, but receives no answer or supporting-fact labels. The recovery model sees only the question, retained prefix, and generic hint; discarded suffix observations are not inserted into its prompt. The same offline collection remains searchable. This is post-hoc selection after failure, not a policy that predicts the origin online. Judge-targeted and reference-informed repair belong only to the archived experiments. Their privileged information and unmatched historical restart hint are documented separately in Appendix B.

4.3 BUDGET SCOPE

The controlled repair cap is the original failed trajectory’s generated-token count. Each request is bounded by the remaining cap, with at most eight new steps and 512 generated tokens per step. Retained steps do not consume new-step opportunities. The historical runner instead counted retained steps toward its eight-step limit and could overshoot its token cap during a final request. Its outcomes remain separate. Even the corrected allowances do not equalize realized tokens, prompt processing, latency or FLOPs; these require additional cost measurements.

5 COMPLETED HOTPOTQA STUDY

5.1 MODEL, COHORT AND HISTORICAL EXCLUSIONS

We evaluated Qwen2.5-32B-Instruct-AWQ on 250 HotpotQA questions (Yang et al., 2018) using the supplied distractor-context paragraphs: 248 questions have ten paragraphs, one has two, and one has six. The offline `search` tool uses exact, substring and fuzzy title matching and returns the first five sentences; `lookup` advances through matching sentences in the current paragraph. Model and tokenizer were pinned to one snapshot (Appendix E). Initial generation was greedy; repairs used temperature 0.7 and seeds 0, 1, 2. Five ordered batches of 50 completed on September 12, 2026. There was no adaptive sample-size change or policy selection after new outcomes.

The cohort was sampled uniformly without replacement from 6,856 sorted eligible IDs using seed 20260912, giving 196 bridge and 54 comparison questions. We excluded 549 distinct IDs: the union of a reconstructed 500-question historical pool and 60 AWS pilot/development questions, with 11 overlaps. The reconstruction reproduces the original sampler, seed and archived raw dataset. The original historical pool JSON was located but its full contents were not downloaded or compared. Thus disjointness is verified against the reconstructed documented history, not every possible undocumented run. This qualification limits any held-out claim.

5.2 SIX CONDITIONS AND SHARED INFORMATION

The treatment is perplexity/argmax with backtracking by two. Four primary controls are full restart, uniform random origin with the same backtracking, position-matched random, and the same uncertainty rule without backtracking. Fixed early origin $\min(1, T - 1)$ is a descriptive sixth condition. All conditions receive the same generic hint, token cap and new-step allowance. The gate uses official HotpotQA answer EM or F1 at least 0.5; strict EM and continuous F1 are secondary outcomes on the same initial-failure cohort. Reference answers affect evaluation and failure selection, not origin choice or the recovery prompt. There is no diagnosis-model condition in this study.

The position control was fitted to 24 failed original development trajectories without using their repair outcomes. It uniformly samples an observed effective origin, normalizes by the development trajectory length and maps to the new length with nearest-position, half-up rounding. All 50 development source questions are excluded. The frozen profile reads no test uncertainty scores; matching in expectation does not guarantee identical finite-sample histograms. Appendix E specifies all origin formulas and profile provenance.

5.3 PAIRED ANALYSIS AND EXECUTION AUDIT

Initial generation succeeded on 137/250 questions (54.8%), with 97 exact matches (38.8%) and mean answer F1 0.5164. The remaining 113 failures each received six conditions and three seeds: 339 trials per condition and 2,034 condition/seed rows. Identical effective origin/prompt/seed executions were shared, yielding 1,058 unique repair executions. The question, not the seed row, is

Table 1: Completed HotpotQA repair comparison on 113 initial failures, with three seeds per condition. Success is answer EM or F1 ≥ 0.5 ; EM and F1 use the same cohort. The final column reports effective origin zero. Fixed early is descriptive. Copied execution rows are not independent trials.

Condition	Successes	Success (%)	EM (%)	F1	$k' = 0$ (%)
Full restart	38/339	11.21	8.55	0.1442	100.00
Random + backtrack 2	36/339	10.62	6.78	0.1402	64.01
Fixed early (descriptive)	39/339	11.50	5.31	0.1369	0.00
Uncertainty + backtrack 2	33/339	9.73	7.08	0.1275	61.95
Position-matched random	33/339	9.73	5.60	0.1265	60.77
Uncertainty, no backtrack	29/339	8.55	4.13	0.1095	15.04

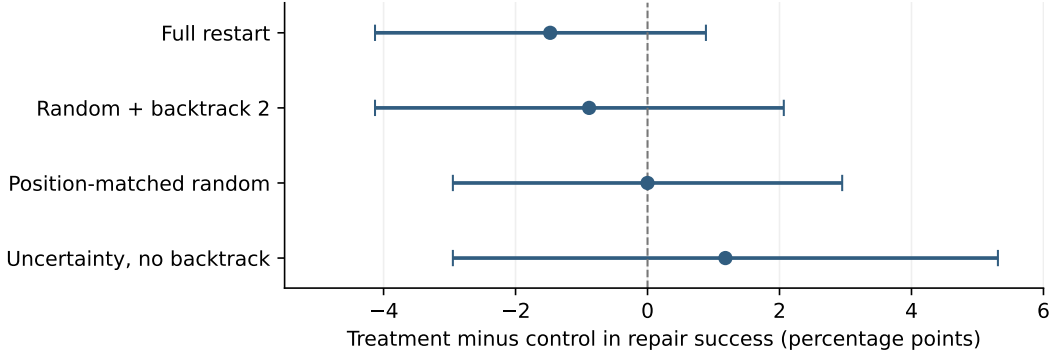


Figure 1: Treatment minus control in repair success. Points are paired question-level differences; bars are individual, unadjusted 95% bootstrap intervals over 113 questions. All four intervals include zero. All four Holm-adjusted sign-flip p -values are 1.000; the exact contrasts and raw p -values appear in Table 9.

the independent resampling unit. We average three seed outcomes within each question and use 10,000 paired question bootstrap resamples for individual 95% intervals. Paired sign-flip tests assume within-question exchangeability. The four prespecified treatment-versus-control comparisons form one Holm family; fixed early is not part of that family or a selected replacement policy.

All five archives passed independent checks against the frozen source, cohort, model and condition manifests. The audit replayed and reference-scored 1,308 trajectories, checked 8,552 observations and 1,742 retained prefix steps, and found zero replay/scoring mismatches or token/new-step violations. The pooled analysis reproduced locally within 10^{-12} of the AWS output. Appendix E reports batch counts, hashes, costs and audit details.

6 CONTROLLED REPAIR RESULTS

Prespecified uncertainty versus controls. The treatment achieves 9.73% repair success (33/339), compared with 11.21% for restart (38/339), 10.62% for matched-offset random (36/339), 9.73% for position-matched random (33/339), and 8.55% for uncertainty without backtracking (29/339). Fixed early achieves 11.50% (39/339), but its descriptive point estimate does not establish a superior policy. Table 1 also reports exact match and F1.

The treatment-minus-restart difference is -1.47 percentage points, with a 95% interval [-4.13, +0.88]. The differences versus offset-matched random, position-matched random and uncertainty without backtracking are -0.88, 0.00 and +1.18 points, with intervals [-4.13, +2.06], [-2.95, +2.95] and [-2.95, +5.31], respectively. No primary comparison shows a clear treatment advantage. These intervals quantify uncertainty for this cohort and setting; they do not establish equivalence, justify a universal rejection of uncertainty signals, or identify a winning alternative from the observed ranking.

How often backtracking becomes restart. The treatment selects effective origin zero in 210/339 rows (61.95%), compared with 64.01% for offset-matched random and 60.77% for the position-matched control. Without backtracking, uncertainty selects zero in 15.04% of rows. Thus many nominal targeted interventions reuse the same physical restart execution. The position control addresses an important alternative explanation, but it matches a development-fitted distribution rather than all features of each individual test trajectory. The observed zero difference does not prove that the policies are equivalent.

Infrastructure cost and measurement scope. Initial generation used 57,499 generated tokens; unique repairs used 303,700 generated and 5,292,100 prompt tokens. Estimated infrastructure usage, including startup, retrieval and storage through the recorded cutoff, was US\$9.16, within a US\$25 session allowance. This is an estimate before credits, tax and transfer, not an invoice or a per-policy efficiency comparison. The instance was confirmed stopped. Equal caps and shared executions do not establish equal realized computation or a wall-clock speedup.

7 DIAGNOSIS AND ALLOWANCE FOLLOW-UP

A separate frozen cohort of 250 HotpotQA questions excludes all 799 previously documented questions. Its 118 initial failures receive full restart, the same uncertainty policy, and a same-model diagnosis/replay adaptation at $0.5\times$, $1\times$, and $2\times$ original generated tokens, with three seeds and eight new steps. The diagnosis sees the question and failed thoughts, actions, and observations, but no reference answer, support labels, or uncertainty scores. It selects a regeneration origin; recovery sees only the retained prefix and common retry hint. This is an untrained adaptation, not a reproduction of Doctor-RAG’s trained system.

At $1\times$, uncertainty succeeds on 35/354 trials (9.89%), versus 37/354 (10.45%) for restart and 36/354 (10.17%) for diagnosis/replay. Uncertainty minus restart is -0.56 percentage points (individual 95% question-bootstrap interval $[-3.67, +2.54]$); versus diagnosis it is -0.28 points ($[-4.24, +3.39]$). All six primary Holm-adjusted p -values across allowances are 1.000. These results provide no clear advantage and do not establish equivalence. Appendix G gives all rates, intervals, costs, and audit counts.

The allowances match recovery-generated tokens; diagnosis adds a measured mean 40.31 generated tokens, 1,234.25 prompt tokens, and one model request per standalone policy attempt. Sharing its acquisition across physical study executions does not make diagnosis free for a deployed attempt.

8 ARCHIVED RESULTS

Older HotpotQA, MuSiQue and 2WikiMultiHopQA summaries cover 854 failures, but retain prompt/budget confounds, retrospective policy selection and unrecovered raw trials. They supply historical context and are not replications of the controlled result. Appendix C contains their rates and diagnostics; these summaries support neither a causal error-cascade claim nor an established length-based recovery rule.

9 ANALYSIS AND DISCUSSION

The following trace, cost and cohort diagnostics were added during manuscript review after the primary results were known. They are exploratory and do not replace the frozen four-comparison family. Their inputs were checked against all five archived batches; Appendix F gives the analysis record, secondary outcomes and illustrative cases.

How much distinct intervention was tested. The treatment and restart share all three executions on 70 of the 113 failed questions. Only 43 questions have a nonzero treatment origin. Let D_q indicate such an origin and let Δ_D denote the mean treatment-minus-restart difference within those questions. Because identical origins, hints and seeds share an execution,

$$\hat{\Delta} = \frac{N_D}{N} \hat{\Delta}_D = \frac{43}{113} (-3.88 \text{ percentage points}) \simeq -1.47 \text{ percentage points.} \quad (5)$$

Table 2: Exploratory execution overlap with the treatment. Shared counts are out of 339 question/seed pairs. Different-question counts are out of 113 questions with any distinct execution. Wins/losses/ties compare three-seed question means, not independent seed trials.

Control	Shared / 339	Different q. / 113	Wins / losses / ties
Full restart	210	43	5 / 8 / 100
Random + backtrack 2	173	68	9 / 9 / 95
Position-matched random	148	93	10 / 10 / 93
Uncertainty, no backtrack	51	96	15 / 10 / 88

The subgroup interval is $[-10.85, +2.33]$ points. This does not change the primary estimand or discard its zero differences: $N = 113$ remains the primary resampling sample. It shows how often the policies can disagree and why the subgroup estimate is imprecise. Across all questions, treatment has a higher seed mean on five, a lower mean on eight, and the same mean on 100. The position-matched comparison has 93 questions with at least one distinct execution, so the lack of a primary advantage cannot be attributed solely to collapse into restart.

Observed resources and unfinished repairs. Uncertainty repair uses 269.9 generated and 4,083.3 prompt tokens per attempt, versus 271.6 and 3,153.3 for restart: 29.5% more prompt tokens with similar output-token use. The paired differences are -1.7 output tokens (individual 95% interval $[-8.9, +5.7]$) and $+930.0$ prompt tokens ($[+579.2, +1321.9]$). Prefix reuse therefore does not imply lower recorded prompt usage. These counts do not measure cached-prefill savings, FLOPs or policy latency; Table 12 reports all six policies.

Original-failure token allowances range from 77 to 548 (median 344), and original failed trajectories have a median of six steps. The treatment ends 103/339 repairs at the token budget and 65/339 at the step limit; together, 168/339 attempts emit no final answer. All 637 original failed-trace steps have usable perplexity, with no empty-profile fallback. Thus missing signal records do not explain the observed result. Stopping at a cap does not identify what the same attempt would achieve with more tokens. Section 7 reports separate executions at three allowances on another frozen cohort.

Metric sensitivity and population interpretation. The strict-EM treatment-minus-restart difference is -1.47 points (95% interval $[-3.83, +0.59]$). Against uncertainty without backtracking, the exploratory EM difference is $+2.95$ points ($[+0.29, +5.60]$); the corresponding primary-gate comparison remains inconclusive. The eight secondary EM/F1 intervals in Table 13 are unadjusted and exploratory. A separate descriptive comparison with fixed early gives -1.77 primary-gate points ($[-6.19, +2.65]$), without adding it to the prespecified family or selecting a replacement policy.

Keeping initially accepted answers and repairing only the 113 rejected ones yields 59.20% full-cohort gate success for treatment and 59.87% for restart. These are reference-gated outcomes, not performance of a deployed failure detector. Forty initially accepted answers fail strict EM and were never repaired; an EM-only failure cohort would need additional executions. This distinction prevents a conditional repair rate from being mistaken for overall answer accuracy.

10 LIMITATIONS

Scope and remaining comparisons. The two completed controlled studies concern one quantized model, separate HotpotQA cohorts, offline evidence, and a permissive answer-success gate. Strict EM and F1 sensitivities retain that initial-failure cohort; an EM-only failure gate would require additional repaired questions. Transfer to other models, live retrieval, and other datasets is not established by these two studies. The separately frozen diagnosis/replay and three-allowance comparison is complete (Section 7); matched runtime and broader replication are separate experiments. Human validation remains necessary for historical judge-localization claims and new mechanism claims; the main repair comparison instead uses deterministic reference-answer scoring.

Exclusions and statistical precision. The test IDs were fixed before new outcomes and are disjoint from declared historical exclusions, but those exclusions depend on reconstruction: the full

original pool was not compared and undocumented runs cannot be ruled out. The position control fits 24 development failures and may imperfectly represent a broader origin distribution. A nonsignificant difference is not evidence of equivalence. Bootstrap intervals are individual intervals; they do not have simultaneous family-wise coverage. Sign-flip inference depends on within-question exchangeability, and seed sharing does not create additional independent questions.

Historical and replay boundaries. The archived summaries retain prompt and budget confounds and lack raw trials for corrected paired inference. They do not inherit the main study’s audit guarantees. Main-study prefix replay was audited against frozen local evidence; this does not guarantee identical stochastic suffixes or restore arbitrary mutable external state. Failure cohorts depend on the model and gate, so their conditional repair rates do not rank population reliability. The infrastructure estimate excludes tax, transfer and subsequent storage, and does not demonstrate matched total inference cost.

11 CONCLUSION

Trajectory repair requires choosing how much prior interaction to retain. In the completed six-condition HotpotQA study, the prespecified uncertainty policy achieved 9.73% repair success versus 11.21% for restart and the same success rate as the position-matched random control. None of the four primary comparisons provides clear evidence of an uncertainty-guided repair advantage. This is a result for one model, environment and allowance, not evidence of equivalence. Exploratory diagnostics show that treatment changes the restart execution on only 43 failed questions and increases mean prompt-token usage by 29.5%, despite similar output-token use. These observations qualify the result’s scope; they do not identify what a larger allowance would achieve on these same questions or how another model would behave. The contribution is a bounded, reproducible empirical comparison. Stronger method, efficiency or generality claims require evidence beyond these conditional recovery comparisons. The separate diagnosis follow-up also found no clear uncertainty advantage across three generated-token allowances.

REPRODUCIBILITY STATEMENT

The controlled study preserves frozen IDs and exclusions, model/source identities, executed notebooks, raw trajectories, repair rows, five archive checksums and independent audit outputs. The downloaded pooled analysis reproduces locally. Appendix E specifies the protocol, provenance limits and audit counts. Review-stage diagnostics supply a checked analysis script, trial and initial-question CSVs, and source/code hashes; their exploratory status is explicit. A local statistical supplement bundles the two CSVs, frozen manifest, reference analyses, checksums and a standalone reproduction script. Extracted into an isolated directory, it reproduces all four primary and eight secondary comparisons, policy costs and full-cohort rates without cloud access. It does not reproduce model generations or replace the raw-trajectory audit. The separate diagnosis follow-up preserves six verified archives, including development, and independent audits of all five main batches. Its local reproduction compares all 3,186 trial rows and 455 numeric values with the remote analysis; the paper’s follow-up tables are generated from that checked output. Paper assets are generated from audited records and separately hashed historical summaries. Full GPU replay packaging and author review before release remain separate; human annotation and unrecovered historical raw trials are not supplied.

AI USE STATEMENT

Language models are used as agents and reference annotators. Generative AI assisted manuscript revision, code inspection, tests, literature lookup and experiment orchestration, artifact auditing and interpretation of results. Authors must verify all assisted content and complete the project-wide disclosure before submission.

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A HISTORICAL SIGNAL AND RULE CATALOG

A.1 SIGNAL DEFINITIONS

The historical framework computes five signal families, oriented so that larger values represent greater uncertainty. Token-based measurements use stored generation log probabilities. Let ℓ_{it} be the log probability of the sampled token t in step i .

Renormalized top- K entropy. For the K_{it} available returned alternatives, define

$$H_{it} = - \sum_{j=1}^{K_{it}} \hat{p}_{itj} \log \hat{p}_{itj}, \quad \hat{p}_{itj} = \frac{\exp(\ell_{itj})}{\sum_{u=1}^{K_{it}} \exp(\ell_{itu})}. \quad (6)$$

The grid uses maximum token entropy, requesting 20 alternatives per token; the returned set can also include the sampled token. This truncated, renormalized distribution does not recover full vocabulary entropy.

Sampled-token probability complement and perplexity. The probability-complement signal is $1 - \exp(\ell_{it})$, aggregated by its maximum within a step. The legacy code calls this metric `max_token_prob`, but the probability belongs to the sampled token. Step perplexity is

$$U_i^{\text{ppl}} = \exp \left(\min \left\{ 20, -\frac{1}{n_i} \sum_{t=1}^{n_i} \ell_{it} \right\} \right). \quad (7)$$

Action disagreement and verbalized uncertainty. Action disagreement resamples the next step five times at temperature 0.7 and computes one minus the fraction matching the most common (action, argument) pair. Verbalized uncertainty is $1 - c_i/100$, where c_i is a confidence score elicited from the model. These signals require additional model calls. Their acquisition cost must be included when comparing total inference efficiency with a restart policy.

A.2 LOCALIZATION RULES

For a profile $U_0, \dots, U_{T-1}$, three standard rules select the largest score, the earliest step among the top three scores, or the earliest step at or above the trajectory’s 75th percentile. Three additional heuristics test upstream and positional preferences:

- **Upstream:** move two scored positions earlier than the uncertainty maximum, clamping to the earliest scored step.
- **Gradient:** select the step with the largest increase in uncertainty between successive scored steps.
- **Position-weighted:** maximize $0.5\tilde{U}_i + 0.5(1 - \tilde{i})$, using min–max normalized uncertainty and position.

Ties favor earlier steps. Missing scores are skipped. Upstream localization and explicit backtracking can compound: selecting two steps before a peak and then backtracking two more may move four positions upstream. These heuristics encode preferences about where to intervene; their use alone does not establish that uncertainty peaks reveal causal error propagation.

The grid contains 30 metric–rule combinations. Crossing them with two backtrack offsets (0, 2) and two hint types yields 120 configurations. Four judge-targeted variants plus random repair and full restart yield 126 configurations. Three seeds are configured for each. Results with the same question, effective origin, hint, and seed can reuse the same generated suffix; their copied result rows are not independent samples.

Table 3: Recorded configuration for the archived QA study. A configured setting is not independent verification of the model actually loaded; runtime logs remain necessary for that check.

Component	Recorded setting
Agent	Qwen2.5-32B-Instruct-AWQ
Reference annotator	Qwen2.5-72B-Instruct-AWQ
Initial questions	500 per QA dataset
Initial generation	Temperature 0; at most 8 steps
Per-step generation limit	512 tokens
Saved token alternatives	Top 20
Repair generation	Temperature 0.7; seeds 0, 1, 2
Nominal recovery cap	Original generated-token count ($m = 1$)
Stored-token signals	Maximum entropy, probability complement, perplexity
Extra-call signals	Five-sample action disagreement; elicited confidence
Repair grid	5 signals, 6 rules, 2 offsets, 2 hint types
Additional configurations	4 reference-targeted variants, random, restart

B EXECUTION AND ANNOTATION DETAILS

B.1 ENVIRONMENT AND PREFIX REPLAY

The agent emits a thought, one action, and its argument. A `search` action queries an entity or title in the question’s offline collection; `lookup` advances through matching sentences in the current page; `finish` supplies the final answer. The model is instructed not to generate the observation itself. The scratchpad contains the question followed by the retained thoughts, actions, arguments, and observations. Recovery replays the retained tool actions in a fresh environment before generating a suffix. It therefore retains both the language-model-visible prefix and the corresponding tool state, rather than merely truncating an answer string.

Replay audit and reproducibility boundary. The corrected runner re-executes retained tool actions and rejects a mismatch in the stored observation, tool-call status or retrieved title before further inference. The main-study independent audit additionally replayed all saved trajectories from frozen records and checked the retained prefix state. It found no mismatches across 8,552 observations and 1,742 retained steps. This audit applies to the new study; historical aggregate results do not acquire the same guarantee. Following the distinction between recorded-prefix reconstruction and live continuation in SymTrace (Luan et al., 2026), prefix fidelity is reported separately from suffix outcomes. Equal numeric seeds do not guarantee identical continuations across different prompts, batch compositions or hardware. Physical reuse of one execution is distinct from rerunning it twice.

Generated-token cost for the original prefix is sunk for a conditional recovery comparison. Re-reading that prefix can still incur prompt cost, and replay can incur tool overhead. These costs should not be erased by excluding the prefix’s previously generated tokens. The saved baseline summaries report recovery generation and tool calls, not this complete accounting. They also do not establish wall-clock speedups, which depend on batching, cache behavior, and hardware.

B.2 PROMPT AND HINT INFORMATION

The agent’s output template is:

```
Thought: <reasoning for this step>
Action: <search, lookup, or finish>
Action Input: <argument>
```

The configured generic hint is: “Your previous attempt at this step may have been wrong. Reconsider carefully.” It is appended to the scratchpad. The historical restart condition omits this hint. The controlled protocol uses identical generic wording for restart and targeted repair.

Informed hints depend on five annotation categories: wrong search query, wrong fact extraction, faulty reasoning, premature or wrong answer, and formatting/tool error. For example, the search

category suggests changing the entity or using more specific search terms, while the extraction category asks the agent to re-read the passage for the requested detail. The hint does not directly insert a replacement step, but its error type comes from a gold-aware reference process. It is therefore privileged even when the answer string is absent from the hint itself.

B.3 REFERENCE ANNOTATION AND FALLBACKS

The annotator receives the question, reference answer, supporting titles, the failed answer, and the numbered trajectory. Supporting-fact text is included when present. A programmatic check records whether required titles were retrieved and can flag missing pages. The judge is asked for a single earliest responsible step, one error class, and a short justification. Its displayed one-based step index is converted to the zero-based trajectory convention.

The implementation prefers a parsed judge response. If parsing fails and the retrieval check identifies a candidate, it uses the last search step as a retrieval-failure reference; otherwise it defaults to step zero and a reasoning-error label. Parsed out-of-range indices are clamped to the trajectory. These are labeling conventions, not evidence that a uniquely causal step exists. The missing raw annotations prevent reporting the fractions of judge, programmatic, default, and clamped labels. A complete validation must retain these sources rather than treating every reference as a successful judge judgment.

B.4 LOCALIZATION AND OFFSET CONVENTIONS

Score ranking breaks ties toward the earlier step. The top-three repair rule selects the earliest index among its three highest-scored positions, not the first index in a descending-score list. The percentile rule uses linear interpolation to form its threshold and selects the earliest qualifying position. The upstream rule moves two positions through the scored-step list, whereas the additional backtrack offset subtracts two trajectory indices. These operations can differ when scores are missing.

The position-weighted rule min-max normalizes uncertainty and position within the scored trajectory and gives each an equal weight:

$$C_i = \frac{1}{2} \frac{U_i - U_{\min}}{U_{\max} - U_{\min}} + \frac{1}{2} \left(1 - \frac{i - i_{\min}}{i_{\max} - i_{\min}} \right). \quad (8)$$

When either range is zero, the corresponding denominator is replaced by one. Empty score profiles fall back to origin zero in the repair selector. These choices create positional preferences even without a calibrated error signal; they motivate fixed-early and position-matched controls.

C DETAILED HISTORICAL RECOVERY ANALYSES

C.1 ARCHIVED STUDY AND AGGREGATE RATES

The historical study sampled 500 questions each from HotpotQA, MuSiQue (Trivedi et al., 2022), and 2WikiMultiHopQA (Ho et al., 2020), with Qwen2.5-32B as agent and Qwen2.5-72B as reference annotator. It contains 226, 425 and 203 failed questions, respectively, for a total of 854. These summaries describe different cohorts and a confounded historical execution regime; they are not pooled with the controlled study.

Full restart exceeds generic judge-targeted repair in the archived HotpotQA and MuSiQue means. On 2WikiMultiHopQA, moving the judge-selected origin two steps earlier changes success from 10.0% to 22.0%, versus 21.2% for restart. These observations motivate direct evaluation of recovery origins; they do not identify a causal error cascade. The retrospective uncertainty maxima include outcome-based selection and, on two datasets, judge-informed hints. They are not held-out policy estimates.

Source-hashed CSVs and indexed notebook displays recover baseline success, generated-token/tool costs and fixed-origin hint comparisons. The archived snapshot lacks the per-question logs needed for valid paired intervals; its obsolete row-wise bootstrap intervals are not reused. Human reference validation also remains unavailable. Appendix C preserves the detailed recovery, cost, hint and localization analyses, and Appendix D explains candidate coverage and snapshot limits.

Table 4: Archived mean repair success (%). Judge-targeted rows use the generic retry hint. The last row is a retrospective maximum over the evaluated uncertainty grid, including informed variants; it is not a held-out policy. Historical budgets require auditing, and valid confidence intervals are pending.

Strategy	HotpotQA	MuSiQue	2WikiMHQA
Random origin	8.1	3.8	19.0
Judge-targeted	10.9	4.9	10.0
Judge-targeted + backtrack 2	12.4	5.6	22.0
Full restart	14.2	5.9	21.2
Retrospective best uncertainty variant	12.8	5.6	25.5

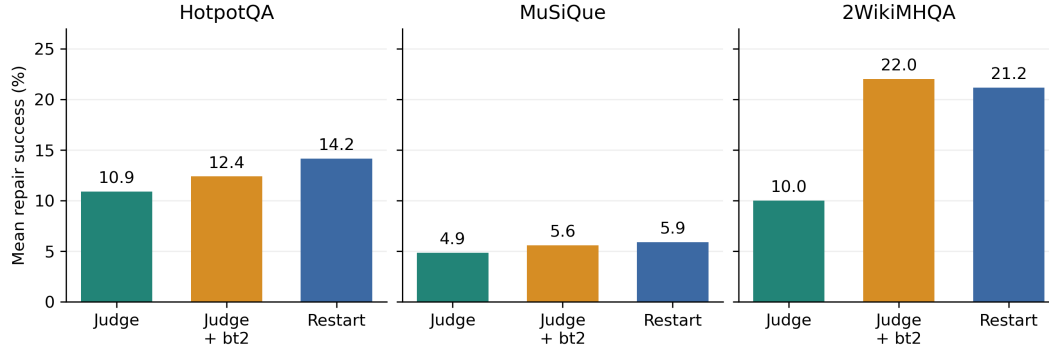


Figure 2: Descriptive archived rates for judge-targeted repair, two-step backtracking, and restart. All bars use mean seed success. The figure does not encode uncertainty intervals or establish a causal mechanism.

C.2 RQ1: TARGETED REPAIR AND RESTART

Full restart has higher archived mean success than judge-targeted repair on HotpotQA (14.2% versus 10.9%) and MuSiQue (5.9% versus 4.9%). It also exceeds the judge-targeted variants with two-step backtracking on those datasets. This indicates that selecting a judge-labeled error does not, by itself, guarantee a better recovery origin than the beginning of the trajectory. It does not establish why restart is better: annotation error, prefix quality, remaining steps, prompt differences, and budget overshoot remain alternative explanations.

The recovered random-origin means are 8.1%, 3.8%, and 19.0% in the same dataset order. These are mean-success summaries, unlike the lower majority-over-seeds values in an earlier cross-dataset table. On 2WikiMultiHopQA, random repair exceeds the unbacktracked judge-targeted rate. Thus the judge-selected point is not even an empirical upper bound on these baseline interventions. Random placement remains an incomplete control for uncertainty: it does not match the latter’s origin distribution or its possible two-step backtracking.

C.3 RQ2: MOVING THE REGENERATION ORIGIN UPSTREAM

Backtracking the judge-selected origin increases archived mean success by 1.5, 0.7, and 12.0 percentage points on HotpotQA, MuSiQue, and 2WikiMultiHopQA. The largest change occurs on 2WikiMultiHopQA: 10.0% to 22.0%, compared with 21.2% for restart. This motivates evaluating earlier recovery origins. It is not a direct test of error cascades, since backtracking simultaneously changes retained context, regeneration length, and whether the judge’s selected step remains relevant.

There are two distinct questions within this comparison. An offset can improve a given localizer by preserving less context; it can also make many localizers operationally identical by clamping them to zero. The aggregate gains cannot distinguish these cases. A per-question analysis should report the selected origin before and after the offset, the fraction of interventions becoming restarts, and

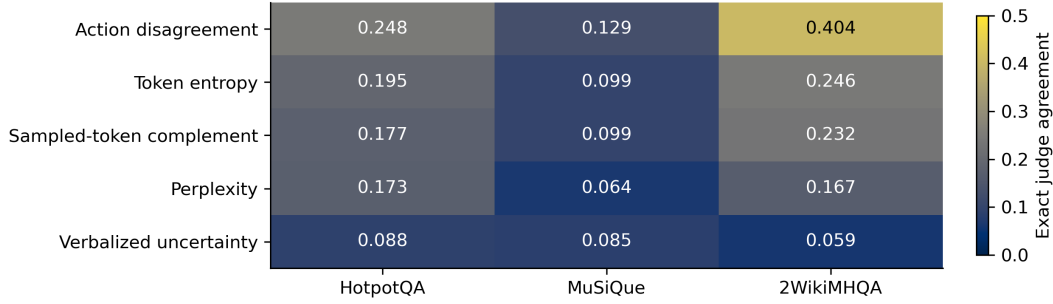


Figure 3: Exact agreement with the reference error step for argmax localization. Larger values indicate annotation agreement, not repair success or calibrated error probabilities. All entries are archived fractions; human validation is not available in the snapshot.

the gain among those retaining a nonempty prefix. These quantities require the original selection records.

C.4 RETROSPECTIVE UNCERTAINTY SELECTION

The aggregate maximum over uncertainty configurations is 12.8% on HotpotQA, 5.6% on MuSiQue, and 25.5% on 2WikiMultiHopQA. The corresponding saved notebook identifies an informed probability-complement/top-three variant on HotpotQA, an informed perplexity/position-weighted/backtrack-two variant on MuSiQue, and a generic perplexity/top-three/backtrack-two variant on 2WikiMultiHopQA. These are selected on the same questions used to report their success. Their rates describe the explored grid, not the expected performance of a policy chosen without looking at test outcomes. In particular, the first two maxima do not isolate uncertainty from judge-provided hints.

C.5 RQ3: LOCALIZATION IS A SEPARATE OUTCOME

Action disagreement has the largest exact agreement among the five reported argmax metrics in each dataset (Figure 3; the numeric table is in Appendix D). The corresponding values are 0.248, 0.129, and 0.404 for HotpotQA, MuSiQue, and 2WikiMultiHopQA. This ordering concerns the reference label, not the success of every metric–rule repair combination. A fair random localization reference must use the eligible steps of each original trajectory, and requires the underlying records. Likewise, a comparison of uncertainty with random repair should match backtracking, hint access, budget, and question coverage, then evaluate a prespecified strategy.

The signal with the greatest annotation agreement need not be the cheapest to acquire: action disagreement requires five additional sampled actions per measured step, whereas stored-token signals reuse the initial rollout. Nor does the highest agreement establish the best downstream policy. The reported repair maximum varies both the signal and the localization rule, sometimes using privileged hints. Comparing that maximum with an argmax localization score would change multiple factors at once.

C.6 RQ4: RECOVERY COST AND HINT INFORMATION

Generated-token and tool costs. Table 5 restores the recorded recovery-only costs for the four baselines. These are rounded means from the same saved displays as their success rates. They exclude the reused prefix from generated-token counts but do not measure prompt processing or all policy overhead. They are therefore descriptive resource measurements, not an audited comparison at equal total computation.

On HotpotQA, two-step-backtracked judge repair uses 272 generated tokens and 4.51 tool calls on average, versus 274 and 4.74 for restart, while its success rate is lower. On MuSiQue the corresponding token means are 347 and 348. Retaining a prefix does not necessarily yield a substantial

Table 5: Archived mean generated recovery tokens and recovery tool calls. HP: HotpotQA; MS: MuSiQue; 2W: 2WikiMultiHopQA. Prompt, uncertainty, diagnosis, and selection costs are not included.

Strategy	HP tokens	HP tools	MS tokens	MS tools	2W tokens	2W tools
Random origin	182	2.80	216	3.90	170	2.36
Judge-targeted	236	3.70	320	5.82	171	2.26
Judge + backtrack 2	272	4.51	347	6.39	234	3.68
Full restart	274	4.74	348	6.62	246	4.03

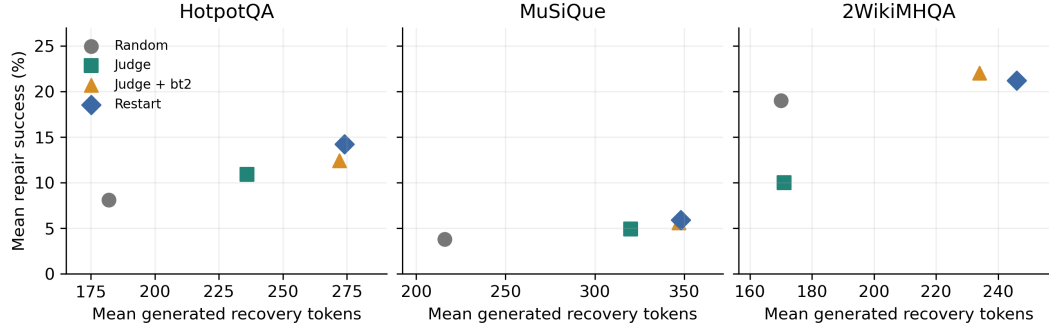


Figure 4: Recovery-token cost versus success for the four recoverable baseline summaries. Markers are historical point estimates, with no confidence intervals. This is not the full-grid Pareto frontier and does not include all inference costs.

generated-token saving. On 2WikiMultiHopQA, backtracked judge repair has both a lower displayed token mean (234 versus 246) and a higher displayed success rate (22.0% versus 21.2%) than restart. This is a point-estimate comparison, not statistically established dominance.

Informed versus generic nudges. An origin-preserving hint comparison must keep backtracking fixed. Table 6 does so for the reference-targeted conditions. At offset zero, informed hints change the displayed rates by +0.5, 0.0, and +2.2 percentage points; at offset two the changes are −0.6, 0.0, and −1.0. Zero denotes equality at one-decimal display precision, not exact equality or statistical equivalence. Keeping origins fixed is necessary to distinguish a hint change from a backtracking change.

The direction varies with the origin, so these summaries support neither a blanket advantage nor a blanket penalty for more specific hints. An informed prompt might focus a useful correction, or direct the model toward an incorrect diagnosis. Distinguishing these explanations requires paired outcomes and diagnosis-quality annotations. In particular, the comparison does not demonstrate a gold-free self-correction capability.

D EXPANDED ARCHIVED DIAGNOSTICS

D.1 CANDIDATE COVERAGE AND RECOVERABLE COSTS

The saved “any uncertainty strategy succeeded” values are 36.0% on HotpotQA, 16.2% on MuSiQue, and 46.0% on 2WikiMultiHopQA. The evaluation code forms this pseudo-strategy by taking the maximum success within each question and seed over the zero-additional-backtrack uncertainty variants, including informed variants. It does not select an answer without the benchmark reference. Also, a rule with zero additional backtracking can still have an upstream preference built into the localizer.

The associated token totals sum nominal member rows. Because several members may select the same origin and hint and reuse one generated suffix, those totals can double-count physical executions. The earlier 47–70-fold compute claim is therefore not retained. These coverage values describe a candidate set under privileged retrospective selection; they should not be placed on an

Table 6: Reference-targeted hint comparison at fixed backtrack offsets. Delta is informed minus generic, in percentage points, calculated from rounded archived means. Informed hints use the gold-aware annotation.

Dataset	Backtrack	Generic (%)	Informed (%)	Delta (pp)
HotpotQA	0	10.9	11.4	+0.5
HotpotQA	2	12.4	11.8	-0.6
MuSiQue	0	4.9	4.9	+0.0
MuSiQue	2	5.6	5.6	+0.0
2WikiMHQA	0	10.0	12.2	+2.2
2WikiMHQA	2	22.0	21.0	-1.0

Table 7: Exact reference-step agreement for argmax, underlying Figure 3. Entries are fractions.

Metric	HotpotQA	MuSiQue	2WikiMHQA
Action disagreement	0.248	0.129	0.404
Token entropy (max)	0.195	0.099	0.246
Sampled-token complement (max)	0.177	0.099	0.232
Perplexity	0.173	0.064	0.167
Verbalized uncertainty	0.088	0.085	0.059

operational Pareto frontier alongside single-output policies. A valid ensemble comparison must freeze a gold-free selector and count unique candidates, acquisition calls, and selection.

D.2 WHY SEED AGGREGATION CHANGES THE QUESTION

For three repair seeds, a question with outcomes $(1, 0, 0)$ contributes $1/3$ to the mean one-attempt repair rate and zero to majority repeatability. A question with $(1, 1, 0)$ contributes $2/3$ and one, respectively. Both summaries can be useful, but mixing them across rows changes the estimand. The restored main table consistently uses mean-success summaries. The earlier notebook’s confidence intervals are not reused because its row-wise bootstrap does not preserve the question-level dependence.

The available displays show only ten strategy rows out of 127, including the coverage pseudo-strategy. The reconstruction parses those visible rows, rejects missing or duplicated required baselines, and checks three baseline rates against the cross-dataset CSVs. It does not infer the hidden rows, reconstruct their joint outcome distribution, or treat rounded costs as exact measurements. This distinction is why the cost plot shows baseline points rather than recreating a full-grid Pareto boundary.

D.3 LEGACY INTERPRETATION LIMITS

Context effects are hypotheses. The possibility that retained errors constrain recovery is consistent with the observed benefit of changing origins. However, the archived “trajectory length” statistic is the average number of tool calls in full-restart repairs. It is a post-intervention outcome, not the original failed trajectory’s length. Correlating four dataset averages cannot justify a four-step routing threshold. Testing the role of prefix context requires original lengths, within-dataset comparisons, and interventions that manipulate the prefix while holding other factors constant.

Error propagation versus recovery opportunity. Backing up a rollout removes later reasoning and observations, but under the historical total-step limit it also grants more new agent turns. Both changes can improve repair without proving that the first retained error caused a downstream uncertainty peak. An error-cascade explanation requires showing the ordering of reference errors and uncertainty peaks, then testing whether the proposed upstream intervention improves recovery beyond a position-matched control. A controlled new-step regime separates some of these factors; the historical total-step regime remains a useful, separately labeled operational setting.

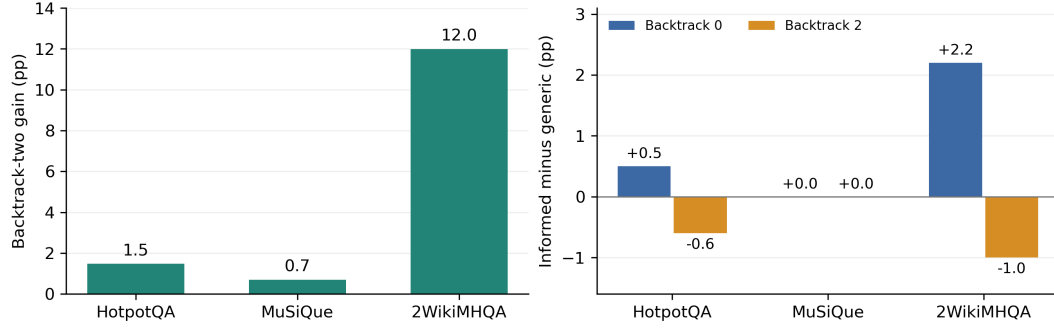


Figure 5: Two descriptive changes in the reference-targeted comparisons. Left: moving the origin two steps earlier, from archived mean-rate CSVs. Right: informed minus generic hints at fixed offsets, from rounded saved notebook displays. Neither panel provides paired confidence intervals.

Failure-mode analysis. Two hypotheses are an error preceding the uncertainty peak and legitimate uncertain exploration. The first requires the proportion of localization misses with $k_J < k_U$, its denominator, and annotation source. The existing exploration flag only identifies missed search/lookup actions, not their correctness. These overlapping hypotheses do not establish the originally asserted 70% and 20% prevalences.

Candidate coverage is not answer selection. An “any strategy succeeded” aggregate uses benchmark answers, so it measures retrospective coverage, not operational ensemble accuracy. Deployment requires a gold-free selector and costs for unique executions and selection; summing nominal strategy costs double-counts shared work.

From diagnosis to a repair policy. A policy must select an origin before observing its repair outcome. Predicting judge agreement does not establish repair utility, and selecting the best test rollout is retrospective. The completed study evaluates six prespecified conditions with matched hints and allowances. An all-origin matrix is deferred; neither an origin-wide optimum nor an all-origin cost frontier is estimated. Appendix E separates completed work from deferred extensions.

The practical implication concerns evaluation: measure recovery directly, include restart, and separate information access from origin selection. The results justify neither a length router, an advantage over diagnosis-and-replay, nor deployment of a gold-selected ensemble.

E COMPLETED CONTROLLED PROTOCOL, AUDIT AND COSTS

This appendix records the completed HotpotQA experiment. The protocol and cohort were frozen before new outcomes, but were not publicly preregistered. Historical results elsewhere in the paper are not relabeled as its outcomes.

E.1 FROZEN CONDITIONS AND MODEL

The main model and tokenizer snapshot was Qwen/Qwen2.5-32B-Instruct-AWQ, revision 5c7cb76a268fc6cfbb9c4777eb24ba6e27f9ee6c. The run used one AWS London g7e.2xlarge with a 96GB RTX PRO 6000 GPU. Initial generation was greedy; repairs used temperature 0.7, seeds 0, 1, 2 and batch size two. Every repair had at most eight new steps, 512 generated tokens per step, and total cap $B_q = G_q$, the original failed trajectory’s generated-token count. The complete failed trace’s uncertainty profile selects the origin; the recovery prompt contains only the retained prefix and generic hint. The discarded suffix and reference labels are not inserted into that prompt.

The position profile contains 24 failed original development trajectories; its effective-origin counts are 14 at zero, two at one, five at three, one at four and two at five. It uses no development repair outcomes. For a uniformly sampled development entry j with origin k_j and length T_j , the new

Table 8: The six frozen conditions actually evaluated. k_U is the perplexity argmax, J a uniform draw over eligible original steps, and K_{dev} the mapped development-profile draw described below. Every condition has the same hint and recovery allowance.

Condition	Effective origin
Full restart	0
Offset-matched random	$\max(0, J - 2), J \sim \text{Unif}\{0, \dots, T - 1\}$
Fixed early (descriptive)	$\min(1, T - 1)$
Uncertainty + backtrack 2	$\max(0, k_U - 2)$
Position-matched random	K_{dev}
Uncertainty, no backtrack	k_U

origin is

$$K_{\text{dev}} = \left\lfloor \frac{k_j}{\max(1, T_j - 1)} (T_q - 1) + \frac{1}{2} \right\rfloor. \quad (9)$$

The profile and seeded sampling were frozen before test outcomes. This is a normalized-position mapping, not a fitted conditional model for every possible trajectory length. All 50 development source questions are excluded. The six conditions and three seeds allow at most $18N$ unique repair jobs for N failures; shared effective executions reduced 2,034 rows to 1,058 jobs.

E.2 COHORT RECONSTRUCTION AND IMMUTABLE EVIDENCE

The historical 500-question pool was reconstructed with the original sampler at git commit 7a73907, seed 42, stratification by level and type, and the archived 7,405-record dataset. Its union with 60 AWS development IDs contains 549 distinct exclusions. The original pool file’s displayed date, size and one visible question ID corroborate the reconstruction; they are not a comparison of its full contents. Undocumented historical runs cannot be excluded. The remaining 6,856 sorted IDs supplied the 250-question cohort, sampled without replacement using seed 20260912.

The freeze was recorded on September 12 at 03:27:45 UTC, before the 04:06:56 EC2 start request and 04:11:45 batch-runner start. The five batches were completed and pooled at 05:25:03 UTC without outcome-based changes. The immutable identities are:

- Study:
e4ea08c8291b0f797d05b15110d762634ec48d57ae53e05aaf4373c805986308.
- Source:
b418bca20be1674d65dc76dc14c42f2b6304eee6856d34a6f083c3d4e6ef4eb9.
- Raw dataset:
a9b4c5906587ac0d20eb83dcb29d0240e268e342c170f7f22988f59f13fa82c3.
- Position profile:
3e4e834cca3857f409d0dae0baf710658825687883bc1ae5eeaf1ebb9c02bb8c.

The local reproducibility record includes `study-freeze.json`, `test-ids-250.json`, `historical-reconstruction.json`, `archive-verification.json` and all five executed batch archives. The source identity above is distinct from the individual ZIP/archive checksums.

E.3 QUESTION-LEVEL INFERENCE AND SECONDARY OUTCOMES

All five batches are pooled with equal question weight after averaging seeds. Individual 95% intervals use 10,000 paired question bootstrap resamples with analysis seed zero. The sign-flip tests assume within-question exchangeability; four primary p -values receive one Holm correction. Fixed early remains descriptive. The intervals themselves are not simultaneous family-wise intervals.

Strict exact match and continuous answer F1 are exploratory sensitivities on the same 113 failures. Their full paired outputs are preserved in `pooled-analysis-local.json`; they are not substituted for the primary gate or its comparison family. Changing to an EM-only failure gate would

Table 9: Prespecified treatment-minus-control comparisons in repair success, in percentage points. All use 113 paired question means. Intervals are individual 95% bootstrap intervals; p -values are paired sign-flip tests.

Control	Delta (pp)	95% interval (pp)	Raw p	Holm p
Full restart	-1.47	[-4.13, +0.88]	0.358	1.000
Random + backtrack 2	-0.88	[-4.13, +2.06]	0.695	1.000
Position-matched random	+0.00	[-2.95, +2.95]	1.000	1.000
Uncertainty, no backtrack	+1.18	[-2.95, +5.31]	0.682	1.000

include additional initially accepted questions for which this experiment did not run repairs. Three seeds per question do not triple the sample size. An ideal reference-gated overall success rate would be $A_0 + (1 - A_0)R_a$ when initially successful answers are left untouched; this identity does not measure a deployed detector’s false alarms or regressions.

E.4 INDEPENDENT ARTIFACT AUDIT

All 180 CPU tests and all 22 plan-only notebook cells passed before launch. Each completed batch was archived with a checksum and independently checked after retrieval. The auditor loaded the frozen source bundle, verified source/input hashes, reconstructed prespecified origins and prompts, checked full question/seed/condition coverage, replayed tool observations, and recomputed answer metrics with the official HotpotQA reference scorer.

Table 10: Completed batch coverage. Violations include replay/scoring mismatches and generated-token/new-step allowance violations; every count is zero.

Batch	Initial	Failed	Unique repairs	Condition/seed rows	Violations
1	50	20	195	360	0
2	50	21	212	378	0
3	50	25	223	450	0
4	50	21	196	378	0
5	50	26	232	468	0

The audit covers 1,308 trajectories, 8,552 observations, 1,742 retained prefix steps, 241 input hashes and 2,034 origin/prompt checks. It found zero mismatches and zero token or new-step violations. Local pooling of the retrieved records agrees with the AWS analysis within 10^{-12} . These checks verify the saved evidence and local tool replay, not bitwise reproduction of stochastic continuations or independent human validation of reference judgments.

E.5 INFRASTRUCTURE USAGE AND COST LIMITS

The run had a US\$25 gross allowance inside the original US\$119 allocation. Guest powerdown was recorded at 05:28:11 UTC. After a retrieval-only restart at 08:48:58, AWS recorded a user-initiated stop reason at 08:54:09 and was observed stopping at 08:55:59. The instance was independently confirmed stopped with no public IP at 19:02:37. No inference was repeated for retrieval.

The cost worksheet uses the verified US\$5.84531/hour rate, approximately 81.25 minutes through main guest powerdown, and a conservative 7.01-minute retrieval interval through the observed stopping state. It adds a one-minute main-transition contingency, retained 200 GiB gp3 storage, and a conservative IPv4 allowance through September 12 at 19:03:19 UTC. The exact provider transition time after main guest powerdown was not captured; the one-minute contingency is an assumption, not a verified upper bound.

The estimate is US\$9.16 before credits, tax and transfer; unrelated account workloads and later storage are excluded. Retained storage continues at approximately US\$0.62/day. Original generation used 57,499 generated tokens and unique repairs used 303,700 generated and 5,292,100 prompt tokens. Acquisition, prefix processing, tool latency, batching and selection must be accounted for to support a policy-efficiency claim; these totals alone do not supply that comparison.

Table 11: Estimated infrastructure usage through the recorded report cutoff. This is not an AWS invoice or a matched-total-compute comparison.

Component	Estimated USD
Main compute through guest powerdown	7.915
Retrieval through observed stopping	0.683
One-minute transition contingency	0.097
Retained storage through report cutoff	0.385
Conservative public IPv4 allowance	0.075
Total before credits, tax and transfer	9.16

E.6 SEPARATE FOLLOW-UP AND DEFERRED SCIENTIFIC WORK

The six-condition study does not include a diagnosis/replay comparator or additional token allowances. These comparisons were subsequently completed on a separate cohort (Section 7). Replication on the other two QA datasets, a second model, matched runtime, a full-origin matrix and new 72B reference annotations are outside these completed studies. FEVER remains excluded because its inspected loader substituted claim text for retrieved evidence. MuSiQue conversion and answer-alias handling require their own provenance/scoring audit before a corrected run.

Historical localization claims still require independent human judgments. The existing proposal samples 50 failed traces per QA dataset independently of repair success, with two annotators blinded to judge labels and outcomes: 150 trajectories and 300 initial judgments. Individual judgments and adjudication would be preserved separately, with ambiguity and multiple-error cases allowed. No such annotations or annotator labor were supplied here. Context interventions and original-length analyses would require separately declared hypotheses; this paper does not infer error-cascade prevalence or a trajectory-length routing threshold from the archived averages.

F REVIEW-STAGE DIAGNOSTICS

These exploratory analyses were added after the primary study completed. They retain all 250 initial questions, 113 failures, six conditions and three seeds. The diagnostic loader verifies 398 input-to-archive comparisons against the five checksummed archives, rescoring original and repaired answers and rejecting missing trials, inconsistent shared executions or allowance violations. Exported trial and initial-question CSVs, the analysis-code hashes and the review-stage analysis plan accompany the diagnostic JSON. The original primary estimates and Holm family are unchanged.

F.1 POLICY COSTS AND TERMINATION

Costs in Table 12 average seeds within each question. Each standalone policy pays for the execution it would use, even if that execution was shared with another condition during this experiment. The study’s infrastructure totals instead deduplicate physical executions. All six policies use stored information and incur zero additional selection-model requests. CPU scoring, log-probability extraction, cache behavior and scheduling overhead are not measured by token totals.

Table 12: Exploratory mean recovery usage per attempted repair, on 113 questions and three seeds. F/B/S are counts terminating with an answer, at the token budget, or at the new-step limit; each row sums to 339. Prompt and output tokens are separate quantities, not an estimate of FLOPs.

Condition	Output tokens	Prompt tokens	Requests	Tool calls	F / B / S
Full restart	271.6	3153.3	5.26	4.54	162 / 99 / 78
Random + backtrack 2	268.4	4179.6	5.15	4.40	171 / 97 / 71
Fixed early (descriptive)	258.1	3720.6	4.82	4.07	193 / 75 / 71
Uncertainty + backtrack 2	269.9	4083.3	5.11	4.35	171 / 103 / 65
Position-matched random	264.5	3885.8	5.01	4.26	178 / 90 / 71
Uncertainty, no backtrack	255.6	5077.3	4.77	3.99	186 / 91 / 62

A repair can reach its token cap on a request that also emits a final answer or reaches the step limit. Consequently, the recorded termination reason and token-cap equality are different diagnostics: treatment has 103 budget-labelled stops but 152/339 executions use their complete token cap. There is no zero-token allowance in this study. Initial failures comprise 61 completed incorrect answers and 52 original step-limit stops. A different allowance requires separate executions; it cannot be reconstructed by reweighting these stopped traces.

F.2 SECONDARY METRICS AND THE ACTUAL FAILURE GATE

Table 13: Exploratory treatment-minus-control differences on the same 113 initial failures. Individual 95% question-bootstrap intervals are not adjusted for the eight secondary comparisons. EM is in percentage points; F1 differences are multiplied by 100. The original primary tests use the answer-success gate, not these metrics.

Control	EM delta [95% CI], pp	100 Δ F1 [95% CI]
Full restart	-1.47 [-3.83, +0.59]	-1.67 [-4.28, +0.64]
Random + backtrack 2	+0.29 [-2.06, +2.65]	-1.27 [-4.14, +1.45]
Position-matched random	+1.47 [-0.59, +3.83]	+0.11 [-2.48, +2.62]
Uncertainty, no backtrack	+2.95 [+0.29, +5.60]	+1.80 [-1.35, +5.07]

Table 14: Outcomes over all 250 questions under the implemented reference gate: preserve initially accepted answers and apply one sampled repair only to the original 113 failures. Three seeds estimate that repair’s mean outcome. Initial rates were 54.80% gate success, 38.80% EM and 0.5164 F1. These are not outcomes of an EM-only rerun or a deployed failure detector.

Recovery condition	Gate success (%)	EM (%)	F1
Full restart	59.87	42.67	0.5597
Random + backtrack 2	59.60	41.87	0.5579
Fixed early (descriptive)	60.00	41.20	0.5564
Uncertainty + backtrack 2	59.20	42.00	0.5522
Position-matched random	59.20	41.33	0.5517
Uncertainty, no backtrack	58.67	40.67	0.5440

The full-cohort calculation retains the same 40 F1-accepted but non-EM original answers for every policy. It never treats the reference gate as a learned detector and never uses majority-over-seeds voting or an oracle best-of-three selector. Question-bootstrap intervals for these descriptive rates are included in the diagnostic JSON.

F.3 ILLUSTRATIVE TRACES AND SELECTION RULE

For each of three outcome strata—higher treatment seed mean, higher restart seed mean, or zero success for both—we display the first sorted question ID and all three seeds. These outcome-selected cases illustrate measured behavior; they do not estimate prevalence, validate diagnoses or establish corrected causal mechanisms. F1 uses each full displayed answer; none of the 18 answers passes strict EM.

Case 1: Higher treatment seed mean. *In 1632, where was one of the most highly acclaimed English architects born?* Reference answer: East Knoyle. The original trace has 8 steps and a 439-token recovery allowance.

Condition	Origin	Gate; F1, seeds 0/1/2	Answers, seeds 0/1/2
Restart	0	0/0/0; 0.00/0.00/0.00	<i>no answer; no answer; no answer</i>
Uncertainty + bt2	3	0/0/1; 0.00/0.00/0.67	<i>no answer; no answer; East Knoyle, Wiltshire, England</i>

Question ID: 5a7619765542992db9473728.

Case 2: Higher restart seed mean. *Was the Metropolitan Life Insurance Company Tower [Met Life Tower] or the 15 Hudson Yards building designed by the firm of Napoleon LeBrun & Sons?* Reference answer: The Metropolitan Life Insurance Company Tower. The original trace has 8 steps and a 458-token recovery allowance.

Condition	Origin	Gate; F1, seeds 0/1/2	Answers, seeds 0/1/2
Restart	0	0/0/1; 0.31/0.32/0.62	The Metropolitan Life Insurance Company Tower was designed by Napoleon LeBrun & Sons, but there is no information to suggest that 15 Hudson Yards was designed by the same firm.; The Metropolitan Life Insurance Company Tower was designed by Napoleon LeBrun & Sons, but there is no information to suggest that 15 Hudson Yards was designed by them.; The Metropolitan Life Insurance Company Tower was designed by Napoleon LeBrun & Sons.
Uncertainty + bt2	4	0/0/0; 0.31/0.00/0.15	The Metropolitan Life Insurance Company Tower was designed by Napoleon LeBrun & Sons, but there is no information to suggest that Napoleon LeBrun & Sons designed the 15 Hudson Yards building.; <i>no answer</i> ; The Met Life Tower was designed by Napoleon LeBrun & Sons, but the design firm for 15 Hudson Yards is not mentioned in the provided information.

Question ID: 5a7322a25542991f9a20c634.

Case 3: Both policies unsuccessful. *Which movie is loosely based off the Brother Grimm's "Iron Henry"?* Reference answer: The Frog Prince. The original trace has 8 steps and a 382-token recovery allowance.

Condition	Origin	Gate; F1, seeds 0/1/2	Answers, seeds 0/1/2
Restart	0	0/0/0; 0.00/0.00/0.00	<i>no answer; no answer; no answer</i>
Uncertainty + bt2	5	0/0/0; 0.00/0.00/0.00	There is no; <i>no answer; no answer</i>

Question ID: 5a733b835542991f9a20c6b2.

F.4 PRECISION AND FURTHER EVIDENCE

The 43-question nonzero-origin comparison has a 95% interval $[-10.85, +2.33]$ percentage points. Its width limits any claim about targeted repair when a prefix is actually retained. For prospective planning, $n \simeq 1.96^2 \hat{\sigma}_d^2 / h^2$ estimates the independent questions needed for an individual 95% interval half-width h , using the observed variance of paired question differences. For a two-point half-width, the four comparisons give approximately 179, 257, 248 and 476 failed questions, in the order of Table 9. These are design sensitivities, not observed power, guaranteed future precision, simultaneous intervals or an outcome-dependent stopping rule for the completed study.

Each additional experiment must declare its scope and affordable sample size before new outcomes. The separate follow-up now evaluates a same-model diagnosis/replay adaptation and three recovery-token allowances (Section 7). Independently generated second-model cohorts would test transportability, while measured complete-policy latency would address resource questions beyond matched recovery-token caps. Human labels are required for new error-localization or mechanism claims, but are not substituted for the deterministic answer scorer in the present primary comparison.

G DIAGNOSIS FOLLOW-UP: COMPLETE RESULTS AND COSTS

The follow-up freezes five batches of 50 new HotpotQA questions before outcomes. Its 250 IDs exclude the 799 documented earlier IDs. Initial generation succeeds on 132 questions (52.8%); 89

are exact matches (35.6%) and mean answer F1 is 0.4980. Each of the 118 failures receives three policies, three seeds, and three allowances, for 3,186 logical rows. The Qwen32B revision, common retry hint, and eight-new-step allowance match the earlier study. Generation uses temperature zero; repairs use 0.7.

Table 15: Separate diagnosis follow-up. Each policy/allowance has 354 seed trials on the same 118 initial failures. Success is answer EM or F1 at least 0.5. EM and F1 retain that failure cohort.

Allowance	Policy	Successes	Success (%)	EM (%)	F1
0.5×	Restart	10/354	2.82	2.54	0.0303
0.5×	Uncertainty, backtrack 2	10/354	2.82	2.54	0.0286
0.5×	Diagnosis/replay	15/354	4.24	3.67	0.0649
1×	Restart	37/354	10.45	7.63	0.1206
1×	Uncertainty, backtrack 2	35/354	9.89	7.06	0.1122
1×	Diagnosis/replay	36/354	10.17	7.91	0.1306
2×	Restart	44/354	12.43	9.32	0.1654
2×	Uncertainty, backtrack 2	39/354	11.02	8.19	0.1500
2×	Diagnosis/replay	39/354	11.02	9.32	0.1515

Table 16: Uncertainty minus control in the diagnosis follow-up. Intervals are individual, unadjusted 95% question-bootstrap intervals (10,000 resamples). All six two-sided question sign-flip tests share one Holm family.

Allowance	Control	Δ (pp)	95% interval (pp)	p	p_{Holm}
0.5×	Restart	+0.00	[-2.26, +2.26]	1.000	1.000
0.5×	Diagnosis/replay	-1.41	[-4.52, +1.41]	0.471	1.000
1×	Restart	-0.56	[-3.67, +2.54]	0.876	1.000
1×	Diagnosis/replay	-0.28	[-4.24, +3.39]	1.000	1.000
2×	Restart	-1.41	[-4.80, +1.41]	0.492	1.000
2×	Diagnosis/replay	+0.00	[-3.67, +3.39]	1.000	1.000

Diagnosis uses the same model with temperature zero, a 512-token maximum, and one deterministic seed. The response supplies an integer origin and a reason. Only the origin controls repair; neither its prose nor discarded suffix evidence is added to the retry prompt. A malformed response would fall back to restart and retain its cost; none of the 118 diagnoses required that fallback. Reference labels determine failure eligibility but do not enter diagnosis or recovery prompts. This is an untrained origin-selection adaptation, not a trained-method reproduction.

Table 17: Mean incremental resource use per standalone recovery attempt, including a complete diagnosis charge for diagnosis/replay. Generated-token allowances apply to recovery; prompt processing and diagnosis add cost.

Allowance	Policy	Generated	Prompt	Requests
0.5×	Restart	151.3	1463.2	3.49
0.5×	Uncertainty, backtrack 2	151.6	2146.9	3.41
0.5×	Diagnosis/replay	188.5	3289.1	4.19
1×	Restart	274.3	3311.8	5.38
1×	Uncertainty, backtrack 2	274.6	4504.8	5.36
1×	Diagnosis/replay	296.1	5371.0	5.84
2×	Restart	295.5	3522.6	5.56
2×	Uncertainty, backtrack 2	295.1	4710.7	5.53
2×	Diagnosis/replay	311.3	5555.6	5.97

Physical execution sharing yields 2,079 unique repairs and 118 diagnoses. Independent audits replay and rescore 2,329 original/repair trajectories, check 13,504 observations, 2,637 retained prefix steps, and 388 input hashes, and find zero mismatches. Unique repairs use 496,780 generated tokens, 7,763,726 prompt tokens, and 9,731 model requests; diagnoses use 4,756 generated tokens, 145,641 prompt tokens, and 118 requests. A finish action counts as a model request, not a retrieval call. The pooled local analysis matches all 3,186 remote trial rows and 455 numeric results within 10^{-12} .

An earlier development audit erroneously counted finish as retrieval; it stopped before any main question. The corrected auditor and a failing-first regression test changed neither the cohort nor its model, prompts, seeds, allowances, or planned comparisons. Failed attempts remain preserved. The conservative infrastructure estimate for all diagnosis attempts and retrievals is \$18.29, excluding tax and transfer. The completed records support token and request accounting, not equal runtime or FLOPs.

H RELATED-WORK COMPARISON AND REQUIRED CONTROLS

This focused literature update uses primary sources available on September 9, 2026; it is not an exhaustive survey. Table 18 compares designs, not repair rates across incompatible cohorts.

Table 18: Closest design overlap and the corresponding evaluation requirement. The last column distinguishes evaluation designs; completion of individual controls is specified in the text. It does not attribute shortcomings to the cited work.

Work	Relevant design	Required distinction here
Doctor-RAG	Trained diagnosis and error-specific QA repair	Compare a diagnosis-based policy; separate origin choice, hints, evidence access, and diagnosis cost.
ReAgent	Local/global rollback in multi-agent QA	Do not claim rollback itself as new; our experiment changes a single agent’s recovery origin.
DoVer	Intervention-driven multi-agent debugging	Recovery, not label agreement alone, is already an established evaluation target.
SymTrace	Recorded-prefix reconstruction and live suffix generation	Audit replay fidelity independently of repair success and suffix randomness.
AgentRewind	Context/workspace restoration with rewind memory	Do not generalize read-only tool replay to external side-effect recovery.
AgentRx; AgenticRAG-FP	Constraint-supported diagnosis; injected-fault attribution	Keep diagnosis evidence and causal identification separate from recovery utility.

Nearest-method comparator. Doctor-RAG’s reasoning-repair operator can reuse documents retrieved throughout the failed trace, and its study uses an EM-only failure gate (Jiao et al., 2026). A direct comparison must reconcile these differences, not copy published rates into our table. Prefer a verified released method when affordable. Otherwise label a same-model diagnosis-and-replay implementation as an adaptation, not a Doctor-RAG reproduction. An origin-only arm should use the same generic hint and prefix-only recovery information as uncertainty. A full-policy arm may use diagnosis-specific guidance, but must declare that access and include diagnosis, tool and selection costs. Report training/distillation costs separately if incurred. The six-condition study implements neither arm. The separate follow-up evaluates the origin-only adaptation with its full acquisition charge; it does not evaluate diagnosis-specific retry guidance or reproduce the trained Doctor-RAG system.

Position-distribution control. The completed HotpotQA experiment includes both offset-matched random and the development-fitted normalized-position control in Appendix E. The latter does not read test uncertainty profiles. Its realized origin-zero frequency is 60.77%, compared with 61.95% for the treatment, and their paired success difference is zero with a 95% interval of $[-2.95, +2.95]$ points. This does not prove exact distributional matching or policy equivalence. The separate diagnosis/replay adaptation provides a closer comparator, while a verified trained-method reproduction remains untested.

Scope of further evidence. The portable notebook’s four-condition default is a feasibility profile; the frozen main run supplied the position profile and six-condition policy. Further conditions require a new frozen design and an affordable allocation before their test outcomes. The completed cohort must not be expanded or reused for outcome-based policy selection and then presented as untouched evaluation. The current results support a bounded empirical comparison; adding citations alone does not resolve the broader empirical novelty question.